

Yichao Cai PhD Candidate in Computer Science

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RESEARCH PROFILE	I study how training objectives shape what machine-learning models learn, and when good predictive performance guarantees recovery of underlying structure . My research combines mathematical theory with controlled experiments, focusing on causal representation learning, multimodal learning, and foundation models.		
RESEARCH INTERESTS	Multimodal learning; vision-language models; identifiable representation learning; learning objectives for foundation models.		
EDUCATION	PhD Candidate in Computer Science <i>Adelaide University</i>	Sep 2023–Present (Expected completion: June 2027)	
	Supervisors: Prof. Javen Qinfeng Shi (principal) and Dr Zhen Zhang Publications: 6 papers across ICML, NeurIPS, ICLR, and ECCV; 3 as first or co-first author , including a NeurIPS Spotlight .		
	M.Sc. in Instrument Science and Technology <i>Wuhan University of Technology</i>	Sep 2016–Jun 2019	
	Advisor: Prof. Xiao Zhou		
	B.Eng. in Measurement and Control Technology and Instrument <i>Wuhan University of Technology</i>	Sep 2012–Jun 2016	
PUBLICATIONS	[1] The Geometric Mechanics of Contrastive Representation Learning: Alignment Potentials, Entropic Dispersion, and Cross-Modal Divergence. Cai, Y., Zhang, Z., Liu, Y., and Shi, J. Q. International Conference on Machine Learning (ICML), 2026. [2] On the Value of Cross-Modal Misalignment in Multimodal Representation Learning. Cai, Y.[*], Liu, Y.[*], Gao, E., Jiang, T., Zhang, Z., van den Hengel, A., and Shi, J. Q. Advances in Neural Information Processing Systems (NeurIPS), 2025. Spotlight. [*]Equal contribution. [3] CLAP: Isolating Content from Style Through Contrastive Learning with Augmented Prompts. Cai, Y., Liu, Y., Zhang, Z., and Shi, J. Q. European Conference on Computer Vision (ECCV), 2024. [4] I Predict Therefore I Am: Is Next Token Prediction Enough to Learn Human-Interpretable Concepts from Data? Liu, Y., Gong, D., Cai, Y., Gao, E., Zhang, Z., Huang, B., Gong, M., van den Hengel, A., and Shi, J. Q. International Conference on Learning Representations (ICLR), 2026. [5] What Makes a Representation Good for Single-Cell Perturbation Prediction? Jiang, W., Liu, Y., Cai, Y., Gao, E., Dong, J., Abbasnejad, E., Yao, L., and Shi, J. Q. International Conference on Machine Learning (ICML), 2026. [6] Boundary Embedding Shaping with Adaptive Contrastive Learning for Graph Structural Disentanglement. Chen, J., Yin, Z., Cai, Y., Liu, Y., Zhang, Z., Gong, D., and Shi, J. Q. International Conference on Machine Learning (ICML), 2026.		
PREPRINTS	[1] An Identifiability Theory of Masked Prediction: Mode Blindness and Mask Schedules. Cai, Y. and Shi, J. Q. arXiv:2608.01383, 2026.		
INVITED TALKS	Understanding Multimodal Contrastive Learning through Two Lenses Online invited talk, <i>Vision & Sensing Lab, Technion</i> Online invited talk, <i>EML, Helmholtz Munich</i>	Jul 2026	

HONORS & AWARDS	D. R. Stranks Travelling Fellowship	Jun 2026
	NeurIPS Scholar Award	Oct 2025
	University of Adelaide Research Scholarship	Sep 2023
	Outstanding Graduate, Wuhan University of Technology	Jun 2019
TEACHING EXPERIENCE	<i>Adelaide University</i> (Formerly the University of Adelaide)	
	Teaching Assistant, <i>Programming for Artificial Intelligence</i>	Semester 2, 2026
	Teaching Assistant, <i>Neural Networks and Deep Learning</i>	Semester 1, 2026
	Guest Lecturer & Head Tutor, <i>Statistical Machine Learning</i> Delivered a guest lecture on vision-language modeling and coordinated the tutor team.	Semester 2, 2025
	Teaching Assistant, <i>Using Machine Learning Tools</i> Developed automated scripts for efficient and consistent assignment marking.	Trimester 2, 2025
	Teaching Assistant, <i>Concepts in AI and Machine Learning</i>	Semester 1, 2025
ACADEMIC SERVICE	Conference reviewer: AAAI 2027; NeurIPS 2026; ICML 2026; ICLR 2026, 2027.	
	Journal reviewer: Transactions on Machine Learning Research (TMLR).	
PROFESSIONAL EXPERIENCE	AI Engineer	Jun 2020–Aug 2022
	<i>Artificial Intelligence Research Institute, Tellhow Software</i>	
	Developed computer-vision systems for automated power-equipment inspection and safety monitoring, contributing to validation, integration, and deployment.	
	Key skills: Computer vision, algorithm validation, error analysis, and system integration.	
	Software Engineer	Jul 2019–Apr 2020
	<i>Huawei Technologies</i>	
VISITING RESEARCH	Developed, tested, and integrated production communication-service components, including feature implementation and production issue diagnosis.	
	Key skills: Software development, testing, interface debugging, system integration, and production troubleshooting.	
	Visiting Student Researcher	May 2018–Sep 2018
TECHNICAL EXPERTISE	<i>California PATH, UC Berkeley</i>	
	Conducted research on visual-perception algorithms for autonomous driving, including model development and experimental evaluation.	
	Methods: identifiability analysis, latent-variable modeling, population-objective analysis, representation geometry, probing, controlled experimentation, and benchmark design.	
	Software: Python, PyTorch, C/C++, Linux, and Git.	
	Languages: Mandarin Chinese (native); English (professional working proficiency).	